

## 교육용 자료 안내

본 자료는 Darun Color Institute 의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

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## Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

# Material Safety Data Sheet (MSDS)

## 1. Chemical Product and Company Identification

| Item            | Information                                            |
|-----------------|--------------------------------------------------------|
| Product Name    | Educational Z-Brown Powder                             |
| Recommended Use | Cosmetic raw material                                  |
| INCI Name       | Iron Oxides (CI 77499, CI 77492), Iron Hydroxide Oxide |

## 2. Hazards Identification

| Category                 | Details                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GHS Classification       | Specific Target Organ Toxicity (Single Exposure) - Category 3 (Respiratory irritation)                                                                                                                                                                                                                                                                                                                                          |
| Hazard Statements        | H335: May cause respiratory irritation                                                                                                                                                                                                                                                                                                                                                                                          |
| Precautionary Statements | P261: Avoid inhalation of dust, fumes, gas, mist, vapor, spray<br>P271: Use only outdoors or in well-ventilated areas<br>P304+P340: If inhaled: Move to fresh air and rest in comfortable position<br>P312: Seek medical advice if you feel unwell<br>P403+P233: Store in well-ventilated place. Keep container tightly closed<br>P405: Store in locked storage<br>P501: Dispose of contents/container according to regulations |

## 3. Composition/Information on Ingredients

*\*Composition ratios are provided as approximate ranges for educational purposes*

| Ingredient Name             | CAS No.    | Approximate Content (%) |
|-----------------------------|------------|-------------------------|
| Iron Oxide (CI 77499)       | 1309-37-1  | 10~25%                  |
| Black Iron Oxide (CI 77492) | 1317-61-9  | 1~5%                    |
| Iron Hydroxide Oxide        | 20344-49-4 | 70~80%                  |

## 4. First Aid Measures

| Route of Exposure | First Aid Procedure                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye Contact       | Seek emergency medical attention. Rinse thoroughly with water for at least 20 minutes. If symptoms develop, seek medical examination.                                      |
| Skin Contact      | Seek medical attention if skin disorder symptoms persist. Remove contaminated clothing and shoes before reuse. Rinse with running water for at least 20 minutes if needed. |
| Inhalation        | If excessive dust or fumes exposure occurs, remove to clean air. Take coughing or other abnormal symptoms seriously and seek medical                                       |

|                   |                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------|
|                   | examination.                                                                                       |
| Ingestion         | Do not induce vomiting. For ingestion, seek immediate medical attention.                           |
| Medical Attention | Ensure medical personnel are aware of substance exposure and take appropriate protective measures. |

## 5. Fire-Fighting Measures

| Item                         | Information                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Alcohol foam, carbon dioxide, water spray, dry sand for small fires                                                                                             |
| Large Fire Method            | Move container away from fire area if safe. Cool containers with large amounts of water even after extinguishing.                                               |
| Specific Hazards             | Container may explode when heated. Some materials can burn but do not ignite easily. Irritating, corrosive, and toxic gases may be generated during combustion. |
| Protective Equipment         | Wear self-contained breathing apparatus and chemical-resistant protective clothing.                                                                             |

## 6. Accidental Release Measures

| Action                   | Description                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prevention               | Prevent dust formation and scattering. Use full or local exhaust ventilation.                                                                                                                          |
| Environmental Protection | Prevent entry into waterways, sewers, basements, and confined spaces. Spillage may cause contamination.                                                                                                |
| Cleanup Methods          | For small spills: Absorb with inert material and place in container. For large spills: Create barriers to prevent spreading. Cover powder spills with plastic sheet to prevent diffusion and keep dry. |

## 7. Handling and Storage

| Category | Requirements                                                                                                                                                                                             |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Avoid dust inhalation. Handle with care. Use full or local exhaust ventilation. Be cautious when opening containers. Follow all protective measures after handling.                                      |
| Storage  | Store in tightly closed container. Store in cool, dry, and well-ventilated place. Store away from incompatible materials. Empty drums should be completely drained and properly sealed before returning. |

## 8. Exposure Controls/Personal Protection

| Control Type | Requirement |
|--------------|-------------|
|--------------|-------------|

|                        |                                                                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exposure Limits        | Iron Oxide (Fe <sub>2</sub> O <sub>3</sub> ): TWA 5 mg/m <sup>3</sup>                                                                                                                         |
| Engineering Controls   | Maintain ventilation during operation to keep air pollution below exposure limits. Ensure work processes meet labor standards.                                                                |
| Respiratory Protection | Wear KOSHA-certified respiratory protection appropriate for exposed particulate matter. For concentrations below 50 mg/m <sup>3</sup> , use appropriate filter-equipped half-mask respirator. |
| Eye Protection         | Use safety goggles. Install emergency eyewash and shower facilities near work area.                                                                                                           |
| Hand Protection        | Wear appropriate chemical-resistant protective gloves to prevent direct contact.                                                                                                              |
| Body Protection        | Wear appropriate chemical-resistant protective clothing to prevent skin exposure.                                                                                                             |

## 9. Physical and Chemical Properties

| Property         | Value                                                    |
|------------------|----------------------------------------------------------|
| Appearance       | Solid (powder) - Brown to black color                    |
| Odor             | Odorless                                                 |
| pH               | No data available                                        |
| Melting Point    | 1565°C (Iron Oxide)                                      |
| Solubility       | Insoluble in water; soluble in acid                      |
| Specific Gravity | Iron Oxide: 5.24, Iron Hydroxide Oxide: 0.97 (estimated) |

## 10. Stability and Reactivity

| Item                    | Information                                                                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical Stability      | Stable at room temperature and normal pressure                                                                                                                                   |
| Conditions to Avoid     | Heat, sparks, flames, and other ignition sources                                                                                                                                 |
| Incompatible Materials  | Oxidizing agents                                                                                                                                                                 |
| Hazardous Decomposition | Container may explode when heated. Materials are non-flammable but can burn under certain conditions. Irritating, corrosive, and toxic gases may be generated during combustion. |

## 11. Toxicological Information

| Effect                      | Description                      |
|-----------------------------|----------------------------------|
| Acute Toxicity (Oral)       | LD50 > 5000 mg/kg (Rat)          |
| Acute Toxicity (Inhalation) | Dust LC50: 5.05 mg/L 4hr (Rat)   |
| Skin Irritation             | OECD 404: No irritation observed |
| Eye Irritation              | OECD 405: No irritation observed |
| Respiratory Sensitization   | No data available                |

|                                           |                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------|
| Skin Sensitization                        | Guinea pig sensitization: Non-sensitizing                                                |
| Carcinogenicity                           | Iron Oxide: Group A4 (ACGIH)                                                             |
| Respiratory Irritation (Single Exposure)  | May cause respiratory irritation                                                         |
| Target Organ Toxicity (Repeated Exposure) | Lung weight increase observed with prolonged exposure. NOAEC $\geq 30$ mg/m <sup>3</sup> |
| Aspiration Hazard                         | No data available                                                                        |

## 12. Ecological Information

| Category            | Information                            |
|---------------------|----------------------------------------|
| Fish Toxicity       | LC96hr $\geq 50000$ mg/L (Danio rerio) |
| Crustacean Toxicity | EC50 $> 100$ mg/L 48hr (Daphnia magna) |
| Bird Toxicity       | No data available                      |
| Persistency         | No data available                      |
| Degradability       | No data available                      |
| Bio-condensability  | No data available                      |
| Soil Mobility       | No data available                      |
| Other Effects       | No data available                      |

## 13. Disposal Considerations

| Item            | Information                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disposal Method | 1) Treat by neutralization, hydrolysis, oxidation, or reduction<br>2) Incinerate at high temperature or treat by high-temperature fusion<br>3) Solidify and dispose |
| Precautions     | Dispose of contents and container according to Waste Management Act regulations                                                                                     |

## 14. Transport Information

| Item                   | Information                              |
|------------------------|------------------------------------------|
| UN Number              | Iron Oxide (spent): UN 1376              |
| Proper Shipping Name   | No UN hazardous materials classification |
| Transport Hazard Class | Not applicable                           |
| Packing Group          | Not applicable                           |
| Marine Pollutant       | Not applicable                           |

## 15. Regulatory Information

| Item             | Information                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regulatory Basis | Prepared in accordance with Korean Occupational Safety and Health Act and relevant regulations. Iron Oxide is a monitored substance (measurement period: dust and fumes) and special health check subject (12-month cycle). |

## 16. Other Information

| Item          | Information                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disclaimer    | The information in this document is based on accurate data and prepared with reference to MSDS from Korea Occupational Safety and Health Agency. However, we do not guarantee the accuracy of results or values obtained when using this material for specific purposes or mixing with other substances. The company assumes no responsibility for personal losses or property damage to buyers or third parties. |
| Prepared by   | Darun Color Institute                                                                                                                                                                                                                                                                                                                                                                                             |
| Document Type | For Educational Purposes Only                                                                                                                                                                                                                                                                                                                                                                                     |